

TRIGGERS - 2520090081

1. Stored Procedure: Get All Employees

```
MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE '2520090081';
Database changed
mysql> CREATE TABLE DEPT (
->   DEPTNO INT PRIMARY KEY,
->   DNAME VARCHAR(20),
->   LOC VARCHAR(20)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE EMP (
->   EMPNO INT PRIMARY KEY,
->   ENAME VARCHAR(20),
->   JOB VARCHAR(20),
->   MGR INT,
->   HIREDATE DATE,
->   SAL DECIMAL(10,2),
->   COMM DECIMAL(10,2),
->   DEPTNO INT,
->   FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
-> );
```

4. Create SALGRADE

```
MySQL 8.0 Command Line Cli x + v
->   EMPNO INT PRIMARY KEY,
->   ENAME VARCHAR(20),
->   JOB VARCHAR(20),
->   MGR INT,
->   HIREDATE DATE,
->   SAL DECIMAL(10,2),
->   COMM DECIMAL(10,2),
->   DEPTNO INT,
->   FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE SALGRADE (
->   GRADE INT PRIMARY KEY,
->   LOSAL DECIMAL(10,2),
->   HISAL DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO DEPT VALUES
-> (10, 'ACCOUNTING', 'NEW YORK'),
-> (20, 'RESEARCH', 'DALLAS'),
-> (30, 'SALES', 'CHICAGO'),
-> (40, 'OPERATIONS', 'BOSTON');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

Insert DEPT data

```
MySQL 8.0 Command Line Cli  ×  +  ▾

mysql> CREATE TABLE DEPT (
  ->     DEPTNO INT PRIMARY KEY,
  ->     DNAME VARCHAR(20),
  ->     LOC VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE EMP (
  ->     EMPNO INT PRIMARY KEY,
  ->     ENAME VARCHAR(20),
  ->     JOB VARCHAR(20),
  ->     MGR INT,
  ->     HIREDATE DATE,
  ->     SAL DECIMAL(10,2),
  ->     COMM DECIMAL(10,2),
  ->     DEPTNO INT,
  ->     FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE SALGRADE (
  ->     GRADE INT PRIMARY KEY,
  ->     LOSAL DECIMAL(10,2),
  ->     HISAL DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO DEPT VALUES
  -> (10, 'ACCOUNTING', 'NEW YORK'),
  -> (20, 'RESEARCH', 'DALLAS'),
  -> (30, 'SALES', 'CHICAGO'),
  -> (40, 'OPERATIONS', 'BOSTON');
Query OK, 4 rows affected (0.01 sec)
```

6. Insert EMP data

```
MySQL 8.0 Command Line Cli  X  +  v

->     GRADE INT PRIMARY KEY,
->     LOSAL DECIMAL(10,2),
->     HISAL DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO DEPT VALUES
-> (10, 'ACCOUNTING', 'NEW YORK'),
-> (20, 'RESEARCH', 'DALLAS'),
-> (30, 'SALES', 'CHICAGO'),
-> (40, 'OPERATIONS', 'BOSTON');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO EMP VALUES
-> (7369, 'SMITH', 'CLERK', 7902, '1980-12-17', 800, NULL, 20),
-> (7499, 'ALLEN', 'SALESMAN', 7698, '1981-02-20', 1600, 300, 30),
-> (7521, 'WARD', 'SALESMAN', 7698, '1981-02-22', 1250, 500, 30),
-> (7566, 'JONES', 'MANAGER', 7839, '1981-04-02', 2975, NULL, 20),
-> (7654, 'MARTIN', 'SALESMAN', 7698, '1981-09-28', 1250, 1400, 30),
-> (7698, 'BLAKE', 'MANAGER', 7839, '1981-05-01', 2850, NULL, 30),
-> (7782, 'CLARK', 'MANAGER', 7839, '1981-06-09', 2450, NULL, 10),
-> (7788, 'SCOTT', 'ANALYST', 7566, '1982-12-09', 3000, NULL, 20),
-> (7839, 'KING', 'PRESIDENT', NULL, '1981-11-17', 5000, NULL, 10),
-> (7844, 'TURNER', 'SALESMAN', 7698, '1981-09-08', 1500, 0, 30),
-> (7876, 'ADAMS', 'CLERK', 7788, '1983-01-12', 1100, NULL, 20),
-> (7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, NULL, 30),
-> (7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, NULL, 20),
-> (7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, NULL, 10);
Query OK, 14 rows affected (0.01 sec)
Records: 14  Duplicates: 0  Warnings: 0

mysql> |
```

7. Insert SALGRADE data

```
mysql> INSERT INTO SALGRADE VALUES
-> (1, 700, 1200),
-> (2, 1201, 1400),
-> (3, 1401, 2000),
-> (4, 2001, 3000),
-> (5, 3001, 9999);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> |
```

8. Verify everything



```
-> (7876, 'ADAMS', 'CLERK', 7788, '1983-01-12', 1100, N
-> (7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, N
-> (7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, N
-> (7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, N
```

Query OK, 14 rows affected (0.01 sec)

Records: 14 Duplicates: 0 Warnings: 0

mysql> INSERT INTO SALGRADE VALUES

```
-> (1, 700, 1200),
-> (2, 1201, 1400),
-> (3, 1401, 2000),
-> (4, 2001, 3000),
-> (5, 3001, 9999);
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> 8. Verify everything

```
-> SHOW TABLES;
```

ERROR 1064 (42000): You have an error in your SQL syntax;
t syntax to use near '8. Verify everything

SHOW TABLES' at line 1

mysql> SHOW TABLES;

```
+-----+
| Tables_in_2520090081 |
+-----+
| dept                  |
| emp                   |
| salgrade              |
| student_marks         |
+-----+
```

4 rows in set (0.01 sec)

mysql> |

```
MySQL 8.0 Command Line Cli x + v
mysql> SHOW TABLES;
+-----+
| Tables_in_2520090081 |
+-----+
| dept
| emp
| salgrade
| student_marks
+-----+
4 rows in set (0.01 sec)

mysql> SELECT * FROM EMP;
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB      | MGR | HIREDATE | SAL      | COMM      | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH | CLERK    | 7902 | 1980-12-17 | 800.00 | NULL      | 20      |
| 7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00    | 30      |
| 7521 | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00    | 30      |
| 7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 2975.00 | NULL      | 20      |
| 7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00   | 30      |
| 7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850.00 | NULL      | 30      |
| 7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450.00 | NULL      | 10      |
| 7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3000.00 | NULL      | 20      |
| 7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL      | 10      |
| 7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 | 0.00      | 30      |
| 7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100.00 | NULL      | 20      |
| 7900 | JAMES | CLERK    | 7698 | 1981-12-03 | 950.00  | NULL      | 30      |
| 7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3000.00 | NULL      | 20      |
| 7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 | NULL      | 10      |
+-----+-----+-----+-----+-----+-----+-----+-----+
14 rows in set (0.00 sec)

mysql> |
```

1. Procedure with IN Parameter

```
MySQL 8.0 Command Line Cli  ×  +  ∨

| FORD | ANALYST | 3000.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetEmployeesByDept(IN p_deptno INT)
-> BEGIN
->     SELECT ENAME, JOB, SAL
->     FROM EMP
->     WHERE DEPTNO = p_deptno;
-> END //
ERROR 1304 (42000): PROCEDURE GetEmployeesByDept already exists
mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GetEmployeesByDept(20);
+-----+-----+-----+
| ENAME | JOB      | SAL      |
+-----+-----+-----+
| SMITH | CLERK    | 800.00   |
| JONES | MANAGER  | 2975.00  |
| SCOTT | ANALYST  | 3000.00  |
| ADAMS | CLERK    | 1100.00  |
| FORD  | ANALYST  | 3000.00  |
+-----+-----+-----+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> |
```

2. Procedure with OUT Parameter

```
MySQL 8.0 Command Line Cli  ×  +  ▾

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetDeptSalaryTotal(
    ->     IN p_deptno INT,
    ->     OUT p_total DECIMAL(10,2)
    -> )
    -> BEGIN
    ->     SELECT SUM(SAL)
    ->     INTO p_total
    ->     FROM EMP
    ->     WHERE DEPTNO = p_deptno;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GetDeptSalaryTotal(30, @total);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT @total;
+-----+
| @total |
+-----+
| 9400.00 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

3. Procedure with IF Logic

```
MySQL 8.0 Command Line Cli  ×  +  ▾

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GiveRaise(
    ->     IN p_empno INT,
    ->     IN p_pct DECIMAL(5,2)
    -> )
    -> BEGIN
    ->     DECLARE v_sal DECIMAL(7,2);
    ->
    ->     SELECT SAL
    ->     INTO v_sal
    ->     FROM EMP
    ->     WHERE EMPNO = p_empno;
    ->
    ->     IF v_sal < 2000 THEN
    ->         UPDATE EMP
    ->         SET SAL = SAL + (SAL * p_pct / 100)
    ->         WHERE EMPNO = p_empno;
    ->     ELSE
    ->         UPDATE EMP
    ->         SET SAL = SAL + (SAL * (p_pct / 2) / 100)
    ->         WHERE EMPNO = p_empno;
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GiveRaise(7369, 10);
Query OK, 1 row affected (0.01 sec)

mysql> |
```

4. Trigger: Prevent Negative Salary

```
MySQL 8.0 Command Line Cli  ×  +  v

Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GiveRaise(7369, 10);
Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER trg_check_salary
-> BEFORE INSERT ON EMP
-> FOR EACH ROW
-> BEGIN
->     IF NEW.SAL < 0 THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Salary cannot be negative';
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO EMP
-> VALUES
-> (9999, 'TEST', 'CLERK', 7902, '2024-01-01', -500, NULL, 20);
ERROR 1644 (45000): Salary cannot be negative
mysql> INSERT INTO EMP
-> VALUES
-> (9999, 'TEST', 'CLERK', 7902, '2024-01-01', 1500, NULL, 20);
Query OK, 1 row affected (0.01 sec)

mysql> |
```

```
MySQL 8.0 Command Line Cli  x  +  v

-> UPDATE EMP
-> SET SAL = SAL + (SAL * p_pct / 100)
-> WHERE EMPNO = p_empno;
-> ELSE
-> UPDATE EMP
-> SET SAL = SAL + (SAL * (p_pct / 2) / 100)
-> WHERE EMPNO = p_empno;
-> END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GiveRaise(7369, 10);
Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER trg_check_salary
-> BEFORE INSERT ON EMP
-> FOR EACH ROW
-> BEGIN
-> IF NEW.SAL < 0 THEN
-> SIGNAL SQLSTATE '45000'
-> SET MESSAGE_TEXT = 'Salary cannot be negative';
-> END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> |
```

5. Trigger: Salary Audit

```
-> (9999, TEST, CLERK, 7902, 2024-01-01, 1500, NULL, 20);
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE EMP_AUDIT (
-> audit_id INT AUTO_INCREMENT PRIMARY KEY,
-> empno INT,
-> old_sal DECIMAL(7,2),
-> new_sal DECIMAL(7,2),
-> changed_on DATETIME
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> |
```

```
mysql>
mysql> CREATE TRIGGER trg_log_sal_change
    -> AFTER UPDATE ON EMP
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.SAL <> NEW.SAL THEN
    ->         INSERT INTO EMP_AUDIT
    ->             (empno, old_sal, new_sal, changed_on)
    ->             VALUES
    ->             (OLD.EMPNO, OLD.SAL, NEW.SAL, NOW());
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql>
mysql> DELIMITER ;
mysql> UPDATE EMP
    -> SET SAL = 3500
    -> WHERE EMPNO = 7369;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```

->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE EMP
-> SET SAL = 3500
-> WHERE EMPNO = 7369;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM EMP_AUDIT;
+-----+-----+-----+-----+-----+
| audit_id | empno | old_sal | new_sal | changed_on |
+-----+-----+-----+-----+-----+
|         1 |    7369 |   880.00 |  3500.00 | 2026-08-25 21:36:02 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |

```

6. Trigger: Prevent President Deletion

```

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER trg_prevent_manager_delete
-> BEFORE DELETE ON EMP
-> FOR EACH ROW
-> BEGIN
->     IF OLD.JOB = 'PRESIDENT' THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Cannot delete the president record';
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;|

```

```

mysql> SELECT *
-> FROM EMP
-> WHERE EMPNO = 7839;
+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB       | MGR | HIREDATE | SAL    | COMM | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+
| 7839  | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL | 10     |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

```

7. Commands to Check Your Procedures

```
mysql> SHOW PROCEDURE STATUS
-> WHERE Db = DATABASE();
```

Db	Name	Type	Definer	Modified	Created	Security_type	Comment
character_set_client	collation_connection	Database Collation					
2520090081	GetAllEmployees	PROCEDURE	root@localhost	2026-08-25 21:17:10	2026-08-25 21:17:10	DEFINER	
cp850	cp850_general_ci		utf8mb4_0900_ai_ci				
2520090081	GetDeptSalaryTotal	PROCEDURE	root@localhost	2026-08-25 21:31:08	2026-08-25 21:31:08	DEFINER	
cp850	cp850_general_ci		utf8mb4_0900_ai_ci				
2520090081	GetEmployeesByDept	PROCEDURE	root@localhost	2026-08-25 21:29:10	2026-08-25 21:29:10	DEFINER	
cp850	cp850_general_ci		utf8mb4_0900_ai_ci				
2520090081	GiveRaise	PROCEDURE	root@localhost	2026-08-25 21:31:45	2026-08-25 21:31:45	DEFINER	
cp850	cp850_general_ci		utf8mb4_0900_ai_ci				

4 rows in set (0.01 sec)

```
mysql> SHOW CREATE PROCEDURE GetAllEmployees;
```

Procedure	sql_mode	character
_set_client	collation_connection	Database Collation
GetAllEmployees	ONLY_FULL_GROUP_BY, STRICT_TRANS_TABLES, NO_ZERO_IN_DATE, NO_ZERO_DATE, ERROR_FOR_DIVISION_BY_ZERO, NO_ENGINE_SUBSTITUTION	CREATE DEFINER='root'@'localhost' PROCEDURE `GetAllEmployees`()

```
BEGIN
  SELECT ENAME, JOB, SAL
  FROM EMP;
END
```

cp850	cp850_general_ci	utf8mb4_0900_ai_ci

1 row in set (0.00 sec)

```
mysql> DROP PROCEDURE IF EXISTS GetAllEmployees;
Query OK, 0 rows affected (0.01 sec)
```

9. Commands to Check Your Triggers

```
mysql> SHOW TRIGGERS;
```

Trigger	Event	Table	Statement	Timing	Created	Definer
sql_mode	character_set_client	collation_connection	Database Collation			
trg_check_salary	INSERT	emp	BEGIN IF NEW.SAL < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Salary cannot be negative'; END IF; END	BEFORE	2026-08-25 21:32:56.72	ONLY_FULL_GROUP_BY, STRICT_TRANS_TABLES, NO_ZERO_IN_DATE, NO_ZERO_DATE, ERROR_FOR_DIVISION_BY_ZERO, NO_ENGINE_SUBSTITUTION
trg_log_sal_change	UPDATE	emp	BEGIN IF OLD.SAL <> NEW.SAL THEN INSERT INTO EMP_AUDIT (empno, old_sal, new_sal, changed_on) VALUES (OLD.EMPNO, OLD.SAL, NEW.SAL, NOW()); END IF; END	AFTER	2026-08-25 21:35:39.42	ONLY_FULL_GROUP_BY, STRICT_TRANS_TABLES, NO_ZERO_IN_DATE, NO_ZERO_DATE, ERROR_FOR_DIVISION_BY_ZERO, NO_ENGINE_SUBSTITUTION
trg_prevent_manager_delete	DELETE	emp	BEGIN IF OLD.JOB = 'PRESIDENT' THEN			

```
MySQL 8.0 Command Line Cli x + v
-----+-----+-----+-----+
| trg_check_salary          | INSERT | emp  | BEGIN
| IF NEW.SAL < 0 THEN
|   SIGNAL SQLSTATE '45000'
|   SET MESSAGE_TEXT = 'Salary cannot be negative';
| END IF;
END
| BEFORE | 2026-08-25 21:32:56.72 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE
S,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_gene
ral_ci | utf8mb4_0900_ai_ci |
| trg_log_sal_change        | UPDATE | emp  | BEGIN
| IF OLD.SAL <> NEW.SAL THEN
|   INSERT INTO EMP_AUDIT
|     (empno, old_sal, new_sal, changed_on)
|   VALUES
|     (OLD.EMPNO, OLD.SAL, NEW.SAL, NOW());
| END IF;
END | AFTER | 2026-08-25 21:35:39.42 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZER
0,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_prevent_manager_delete | DELETE | emp  | BEGIN
| IF OLD.JOB = 'PRESIDENT' THEN
|   SIGNAL SQLSTATE '45000'
|   SET MESSAGE_TEXT = 'Cannot delete the president record';
| END IF;
END
| BEFORE | 2026-08-25 21:37:19.13 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,N
O_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb
4_0900_ai_ci |
-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

```
-----+-----+-----+-----+
| trg_check_salary          | INSERT | emp  | BEGIN
| IF NEW.SAL < 0 THEN
|   SIGNAL SQLSTATE '45000'
|   SET MESSAGE_TEXT = 'Salary cannot be negative';
| END IF;
END
| BEFORE | 2026-08-25 21:32:56.72 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE
S,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_gene
ral_ci | utf8mb4_0900_ai_ci |
| trg_log_sal_change        | UPDATE | emp  | BEGIN
| IF OLD.SAL <> NEW.SAL THEN
|   INSERT INTO EMP_AUDIT
|     (empno, old_sal, new_sal, changed_on)
|   VALUES
|     (OLD.EMPNO, OLD.SAL, NEW.SAL, NOW());
| END IF;
END | AFTER | 2026-08-25 21:35:39.42 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZER
0,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |
| trg_prevent_manager_delete | DELETE | emp  | BEGIN
| IF OLD.JOB = 'PRESIDENT' THEN
|   SIGNAL SQLSTATE '45000'
|   SET MESSAGE_TEXT = 'Cannot delete the president record';
| END IF;
END
| BEFORE | 2026-08-25 21:37:19.13 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,N
O_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb
4_0900_ai_ci |
-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

```
mysql> SHOW TRIGGERS
-> WHERE 'Table' = 'EMP';
```

Trigger	Event	Table	Statement	Timing	Created	Definition
trg_check_salary	INSERT	emp	BEGIN IF NEW.SAL < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Salary cannot be negative'; END IF;	BEFORE	2026-08-25 21:32:56.72	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION root@localhost cp850 cp850_general_ci utf8mb4_0900_ai_ci
trg_log_sal_change	UPDATE	emp	BEGIN IF OLD.SAL <> NEW.SAL THEN INSERT INTO EMP_AUDIT (empno, old_sal, new_sal, changed_on) VALUES (OLD.EMPNO, OLD.SAL, NEW.SAL, NOW()); END IF;	AFTER	2026-08-25 21:35:39.42	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION root@localhost cp850 cp850_general_ci utf8mb4_0900_ai_ci
trg_prevent_manager_delete	DELETE	emp	BEGIN			